

Topic 4: Electrochemistry - Answers

IGCSE Chemistry 0620 Paper 4 topical answer booklet.

Subtopic: Electrolysis and electrode reactions

Questions in this answer PDF: 18

Use this with the matching topical question PDFs. The answer points are organised by the same source labels.

Short explanation notes are added for harder calculation, electrolysis, equilibrium, rate, salts, organic, metals and practical questions.

Periodic table reference pages are not included to save printing paper.

2020 Feb/March Paper 42 Question 2

Main topic: Topic 4: Electrochemistry | Marks: 19 | Answer source: 0620_m20_ms_42.pdf

Answer points

2(a)(i)

- bauxite

2(a)(ii)

- breakdown by (the passage of) electricity (1)
- of an ionic compound in molten / aqueous (state) (1)

2(b)(i)

- cryolite

2(b)(ii)

- less CO₂ emission

2(b)(iii)

- $\text{Al}^{3+} + 3\text{e}^- \rightarrow \text{Al}$
- any positive Al species gaining electron(s) (1)
- correct species and balance (1)
- $2\text{O}^{2-} \rightarrow \text{O}_2 + 4\text{e}^-$
- any negative O species losing electron(s) (1)
- correct species and balance (1)

2(b)(iv)

- anodes or carbon / graphite react with oxygen / O₂ (1)
- (form) carbon dioxide (1)

2(c)(i)

- amphoteric

2(c)(ii)

- aluminium sulfate (1) $\text{Al}_2(\text{SO}_4)_3$ (1)

2(c)(iii)

- water

2(c)(iv)

- $2\text{Al}(\text{OH})_3 \rightarrow \text{Al}_2\text{O}_3 + 3\text{H}_2\text{O}$
- species (1) balance (1)

2(c)(v)

- aluminium carbonate (1)
- aluminium nitrate (1)

Easy explanation / reminder

- *Electrolysis tip: cations go to the cathode and gain electrons. Anions go to the anode and lose electrons.*

- *Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.*

2020 May/June Paper 42 Question 5

Main topic: Topic 4: Electrochemistry | Marks: 8 | Answer source: 0620_s20_ms_42.pdf

Answer points

5(a)

- breakdown of an ionic compound when molten or in aqueous solution (1)
- (using) electricity / electric current / electrical energy (1)

5(b)

- platinum / graphite

5(c)

- $2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2$

5(d)

- $\text{Na}^+ \text{H}^+ \text{Cl}^- \text{OH}^-$

- all four (2) 3 or 2 (1)

5(e)

- H^+ and Cl^- are discharged / removed (1)

- Na^+ and OH^- remain (1)

Easy explanation / reminder

- *Electrolysis tip: cations go to the cathode and gain electrons. Anions go to the anode and lose electrons.*

- *Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.*

2020 May/June Paper 43 Question 7

Main topic: Topic 4: Electrochemistry | Marks: 14 | Answer source: 0620_s20_ms_43.pdf

Extra topic(s): Topic 9: Metals

Answer points**7(a)**

- breakdown of a molten / or aqueous ionic compound by the passage of electricity

7(b)

- bauxite

7(c)(i)

- it is above carbon in the reactivity series / more reactive than carbon

7(c)(ii)

- any one from:

- -

- aluminium oxide has high melting point / cryolite has lower melting point than aluminium oxide

- -

- using cryolite reduces costs / expensive to melt aluminium

7(c)(iii)

- oxygen

7(c)(iv)

- $\text{Al}^{3+} + 3\text{e}^- \rightarrow \text{Al}$

7(d)

- any two related to use as electricity cables:

- -

- ductile / malleable

- -

- conducts (electricity)

- -

- low density

- -

- protective oxide layer

7(e)(i)

- iron + water + oxygen $\rightarrow$ (hydrated) iron oxide

7(e)(ii)

- any two from:

- -

- act as catalysts

- -

- variable oxidation numbers

- -

- form coloured compounds / coloured ions

- -

- higher melting point

- -

- higher density

- -

- harder

Easy explanation / reminder

- *Rate tip: use collision theory. Faster rate means more frequent successful collisions, or more particles with energy greater than activation energy.*
- *Electrolysis tip: cations go to the cathode and gain electrons. Anions go to the anode and lose electrons.*

2020 Oct/Nov Paper 41 Question 5

Main topic: Topic 4: Electrochemistry | Marks: 15 | Answer source: 0620_w20_ms_41.pdf

Extra topic(s): Topic 7: Acids, bases and salts

Answer points

5(a)(i)

- breakdown by (the passage of) electricity (1)
- of an ionic compound in molten/aqueous (state) (1)

5(a)(ii)

- they do not react

5(a)(iii)

- negative electrode: hydrogen (gas) (1)
- positive electrode: oxygen (gas) (1)

5(a)(iv)

- $\text{H}^+ + \text{e}^-$ as the only species on the left (1)
- equation fully correct (1)
- $2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2$ (scores 2)

5(b)(i)

- $\text{S} + \text{O}_2 \rightarrow \text{SO}_2$

5(b)(ii)

- rate of forward reaction is equal to rate of reverse reaction (1)
- constant concentration (of reactants and products) (1)

5(b)(iii)

- exothermic / heat / energy is released / surroundings warm up
- products have lower energy than reactants / opposite reasoning also allowed

5(c)

- water / H_2O

5(d)

- ($M_r =$) 98
- ($0.75 \times 98 =$) 73.5

Easy explanation / reminder

- *Calculation tip: start with moles = mass / M_r or moles = concentration x volume. Then use the mole ratio from the balanced equation before converting to the final answer.*
- *Equilibrium tip: explain which reaction is favoured and why. A closed system is needed because reactants and products must not escape.*

2021 May/June Paper 41 Question 3

Main topic: Topic 4: Electrochemistry | Marks: 19 | Answer source: 0620_s21_ms_41.pdf

Answer points

3(a)

- $2\text{K} + \text{Cl}_2 \rightarrow 2\text{KCl}$
- Cl_2 on left hand side (1)
- equation fully correct (1)

3(b)

- K outer shell with 8 crosses (1)
- Cl outer shell with 7 dots and 1 cross (1)
- + and - (1)

3(c)(i)

- breakdown by (the passage of) electricity (1)
- of an ionic compound in molten or aqueous (state) (1)

3(c)(ii)

- (anode) chlorine
- (cathode)potassium

3(d)(i)

- $2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2$

- H^+ and e^- on left hand side (1)
- equation fully correct (1)

3(d)(iii)

- chlorine

3(d)(iii)

- potassium hydroxide (1)

3(e)

- one shared pair of electrons and 6 non-bonding electrons on each chlorine atom

3(f)(i)

- liquid (1)
- BOTH melting point is below -75°C AND boiling point is above -75°C
- OR
- BOTH -75°C is higher than -101°C / melting point AND lower than -35°C / boiling point
- OR
- -75°C is between melting point or -101°C and boiling point or -35°C

3(f)(ii)

- ionic bonds in KCl (1)
- attraction between molecules in Cl_2 (1)
- weaker attraction (between particles) in Cl_2 opposite reasoning also allowed (1)

Easy explanation / reminder

- *Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation before converting to the final answer.*
- *Electrolysis tip: cations go to the cathode and gain electrons. Anions go to the anode and lose electrons.*

2021 May/June Paper 42 Question 4

Main topic: Topic 4: Electrochemistry | Marks: 12 | Answer source: 0620_s21_ms_42.pdf

Answer points

4(a)

- electrolyte

4(b)(i)

- $4\text{OH}^- \rightarrow 2\text{H}_2\text{O} + \text{O}_2 + 4\text{e}^-$
- balance of charge (1)
- rest of equation (1)

4(b)(ii)

- (OH^- (aq) ions) lose electrons

4(c)

- fizzing

4(d)

- $2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2$
- species correct (1)
- fully correct equation (1)

4(e)(i)

- fizzing (1)
- green gas (1)

4(e)(ii)

- (litmus turns) blue and alkali / base forms (1)
- Sodium hydroxide / NaOH (forming) (1)

4(f)

- platinum

Easy explanation / reminder

- *Electrolysis tip: cations go to the cathode and gain electrons. Anions go to the anode and lose electrons.*
- *Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.*

2021 May/June Paper 43 Question 3

Main topic: Topic 4: Electrochemistry | Marks: 18 | Answer source: 0620_s21_ms_43.pdf

Answer points

3(a)

- $2\text{Na} + \text{F}_2 \rightarrow 2\text{NaF}$

- F₂ (1)
- equation fully correct (1)

3(b)

- Na outer shell with 8 crosses (1)
- F outer shell with 7 dots and 1 cross (1)
- Na⁺ and F⁻ (1)

3(c)(i)

- breakdown by (the passage of) electricity (1)
- of an ionic compound in molten / aqueous (state) (1)
- Question Answer

3(c)(ii)

- oxygen
- hydrogen

3(d)(i)

- (anode) fluorine (1)
- (cathode) sodium (1)

3(d)(ii)

- Na⁺ + e⁻ → Na

3(e)

- one shared pair of electrons and 6 non-bonding electrons on each fluorine atom

3(f)(i)

- liquid (1)
- BOTH melting point is below -195 °C and boiling point is above -195 °C
- OR -195 °C is in between melting point and boiling point / -220 °C and -188 °C (1)
- OR
- BOTH -195 °C is higher than -220 °C / melting point AND lower than -188 °C / boiling point

3(f)(ii)

- ionic bonds in NaF (1)
- attraction between molecules in F₂ (1)
- weaker attraction (between particles) in F₂ opposite reasoning also allowed (1)
- Question Answer

Easy explanation / reminder

- *Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation before converting to the final answer.*
- *Electrolysis tip: cations go to the cathode and gain electrons. Anions go to the anode and lose electrons.*

2021 Oct/Nov Paper 41 Question 3

Main topic: Topic 4: Electrochemistry | Marks: 13 | Answer source: 0620_w21_ms_41.pdf

Answer points

3(a)(i)

- 2Cl⁻ → Cl₂ + 2e⁻
- Cl₂ (1)
- rest of equation (1)

3(a)(ii)

- Oxidation AND lose electrons

3(b)

- effervescence (of colourless gas)

3(c)

- 2H⁺ + 2e⁻ → H₂
- H⁺ + e as only species on LHS (1)
- rest of equation fully correct (1)

3(d)(i)

3(d)(ii)

- increase (1)
- H⁺ ions being removed (1)

3(e)

- cathode: silver / grey solid (1)
- anode: bubbles of orange / brown gas (1)

3(f)

- inert (1)
- conducts electricity (1)

Easy explanation / reminder

- *Electrolysis tip: cations go to the cathode and gain electrons. Anions go to the anode and lose electrons.*
- *Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.*

2021 Oct/Nov Paper 43 Question 2

Main topic: Topic 4: Electrochemistry | Marks: 12 | Answer source: 0620_w21_ms_43.pdf

Answer points

2(a)

- ionic compound
- AND either molten or aqueous(or both)(1)
- conducts electricity / undergoes electrolysis(1)

2(b)(i)

- oxygen (1)
- pink / brown solid (1)
- copper (1)
- orange / brown / yellow liquid (1)
- bromine (1)

2(b)(ii)

- $2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2$ (2)

2(b)(iii)

- inert (1)
- good conductor of electricity (1)

2(b)(iv)

- electron

Easy explanation / reminder

- *Electrolysis tip: cations go to the cathode and gain electrons. Anions go to the anode and lose electrons.*
- *Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.*

2022 Feb/March Paper 42 Question 4

Main topic: Topic 4: Electrochemistry | Marks: 13 | Answer source: 0620_m22_ms_42.pdf

Extra topic(s): Topic 9: Metals

Answer points

4(a)(i)

- arrow going from Zn to Fe

4(a)(ii)

- $\text{Zn} \rightarrow \text{Zn}^{2+} + 2\text{e}^-$
- Zn as only reactant and Zn^{2+} as only product
- correct equation

4(b)(i)

- any metal above zinc in reactivity series

4(b)(ii)

- any metal below iron in reactivity series

4(c)(i)

- hydrogen and oxygen

4(c)(ii)

- water

4(d)(i)

- electrolysis

4(d)(ii)

- mobile ions

4(e)(i)

- hydrogen
- chlorine
- sodium hydroxide

4(e)(ii)

- sodium
- February/March 2022

Easy explanation / reminder

- *Electrolysis tip: cations go to the cathode and gain electrons. Anions go to the anode and lose electrons.*

2022 May/June Paper 43 Question 5

Main topic: Topic 4: Electrochemistry | Marks: 18 | Answer source: 0620_s22_ms_43.pdf

Extra topic(s): Topic 3: Stoichiometry, Topic 7: Acids, bases and salts

Answer points

5(a)

- (lattice of) positive ions (1)
- sea of / delocalised / mobile electrons (1)
- attraction between positive ions and electrons (1)

5(b)

- copper (1)
- spoon (1)
- (aqueous or solution) of named copper salt (1)

5(c)

- $(0.2 \rightarrow 50 / 1000) = 0.01$ (1)
- 0.01 (1)
- $(250 \rightarrow 0.01) = 2.5$ (1)

5(d)

- solid undissolved

5(e)

- particles have less energy
- fewer collisions (between particles) occur per second / per unit time
- a smaller percentage / proportion / fraction of collisions (of particles) are successful / have energy above activation
- energy / have energy equal to activation energy

5(f)

- copper(II) carbonate OR copper(II) hydroxide

5(g)

- filtration

5(h)

- (a solution that) can dissolve no more solute
- at a given temperature

5(i)

- forms anhydrous (copper sulfate)
- OR forms (white) powder

Easy explanation / reminder

- *Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation before converting to the final answer.*

- *Rate tip: use collision theory. Faster rate means more frequent successful collisions, or more particles with energy greater than activation energy.*

2022 Oct/Nov Paper 43 Question 3

Main topic: Topic 4: Electrochemistry | Marks: 15 | Answer source: 0620_w22_ms_43.pdf

Answer points

3(a)

- bauxite

3(b)

- breakdown by (the passage of) electricity (1)
- of an ionic compound in molten or aqueous (state) (1)

3(c)(i)

- improves conductivity of the electrolyte / makes the electrolyte a better conductor (1)
- lower operating temperature (1)

3(c)(ii)

- $\text{Al}^{3+} + 3\text{e}^- \rightarrow \text{Al}$
- only $\text{Al}^{3+} + (3) \text{e}^-$ on the left (1)

- equation fully correct (1)

3(c)(iii)

- anodes or carbon react(s) with oxygen (1)

- form carbon dioxide (1)

3(d)

- unreactive coating of aluminium oxide (1)

3(e)(i)

- neutralises both acids AND alkalis

3(e)(ii)

- $2\text{NaOH} + \text{Al}_2\text{O}_3 \rightarrow 2\text{NaAlO}_2 + \text{H}_2\text{O}$

- NaAlO_2 on the right hand side (1)

- equation fully correct (1)

3(f)

- GaCl_3 (1)

- $\text{Ga}_2(\text{SO}_4)_3$ (1)

Easy explanation / reminder

- *Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation before converting to the final answer.*

- *Electrolysis tip: cations go to the cathode and gain electrons. Anions go to the anode and lose electrons.*

2023 May/June Paper 41 Question 5

Main topic: Topic 4: Electrochemistry | Marks: 16 | Answer source: 0620_s23_ms_41.pdf

Answer points

5(a)(i)

- breakdown by (the passage of) electricity(1)

- of an ionic compound in molten or aqueous (state) (1)

5(a)(ii)

- graphite is inert AND graphite conducts electricity

5(a)(iii)

- $2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2$

- $\text{H}^+ + \text{e}^-$ as only species on LHS(1)

- equation correct(1)

5(a)(iv)

- electrons

5(a)(v)

- ions

5(a)(vi)

- oxygen(1)

- hydrogen(1)

5(b)(i)

- aluminium oxide

5(b)(ii)

- any two from:

- ■■

- solvent

- ■■

- lowers the operating temperature

- ■■

- increases conductivity

5(b)(iii)

- carbon reacts with oxygen and forms carbon dioxide

5(c)(i)

- $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$

- all formulae(1) equation correct(1)

5(c)(ii)

- no carbon dioxide evolved

- OR

- more efficient

Easy explanation / reminder

- *Electrolysis tip: cations go to the cathode and gain electrons. Anions go to the anode and lose electrons.*
- *Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.*

2023 May/June Paper 43 Question 5

Main topic: Topic 4: Electrochemistry | Marks: 16 | Answer source: 0620_s23_ms_43.pdf

Answer points

5(a)(i)

- ionic compound
- molten and / or aqueous

5(a)(ii)

- oxidation number (of copper)

5(a)(iii)

- fades / (becomes) colourless

5(a)(iv)

- $\text{Cu}^{2+} + 2\text{e}^- \rightarrow \text{Cu}$
- Cu^{2+} and (any number of) e on left hand side
- equation correct

5(a)(v)

- OH^-

5(b)

- anode dissolves

5(c)(i)

- silver
- spoon
- (aqueous or solution) of silver nitrate

5(c)(ii)

- prevent corrosion
- improve appearance

5(d)(i)

- carbon dioxide: (increased) global warming
- carbon monoxide: toxic

5(d)(ii)

- needs high pressure to store hydrogen

Easy explanation / reminder

- *Electrolysis tip: cations go to the cathode and gain electrons. Anions go to the anode and lose electrons.*
- *Metals tip: link the method to the reactivity series. Very reactive metals need electrolysis; less reactive metal oxides can often be reduced by carbon/CO.*

2024 Oct/Nov Paper 42 Question 2

Main topic: Topic 4: Electrochemistry | Marks: 13 | Answer source: 0620_w24_ms_42.pdf

Answer points

2(a)

- bauxite

2(b)

- cryolite

2(c)

- (it has) mobile ions

2(d)(i)

- 4
- $\text{Al}^{3+} + 3\text{e}^- \rightarrow \text{Al}$
- Al^{3+} AND e^-
- correct equation

2(d)(ii)

- electrons are lost (from oxide ions)

2(d)(iii)

- anode / electrode made of carbon or graphite
- anode / electrode reacts with oxygen / burns / combusts (in oxygen)

2(e)

- aluminium is resistant to corrosion

2(f)

- eight crosses in second shell of Al
- seven dots and one cross in second shell of each F
- '3+' charge on Al ion on correct answer line AND '-' charge on each F ion on correct answer line

Easy explanation / reminder

- *Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation before converting to the final answer.*
- *Electrolysis tip: cations go to the cathode and gain electrons. Anions go to the anode and lose electrons.*

2024 Oct/Nov Paper 43 Question 2

Main topic: Topic 4: Electrochemistry | Marks: 11 | Answer source: 0620_w24_ms_43.pdf

Answer points**2(a)**

- breakdown by (the passage of) electricity
- of an ionic compound in molten or aqueous (state)

2(b)(i)

- brown solution OR black solid
- iodine
- hydrogen
- copper

2(b)(ii)

- $4\text{OH}^- \rightarrow 2\text{H}_2\text{O} + \text{O}_2 + 4\text{e}^-$
- OH^- AND e^-
- correct equation

2(c)(i)

- copper OR anode forms Cu^{2+}
- which goes into the solution

2(c)(ii)

- copper

2(c)(iii)

- no change

Easy explanation / reminder

- *Electrolysis tip: cations go to the cathode and gain electrons. Anions go to the anode and lose electrons.*
- *Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.*

2025 May/June Paper 42 Question 3

Main topic: Topic 4: Electrochemistry | Marks: 15 | Answer source: 0620_s25_ms_42.pdf

Answer points**3(a)**

- ionic
- molten
- in aqueous solution

3(b)(i)

- increase

3(b)(ii)

- becomes paler blue
- or
- becomes colourless

3(b)(iii)

- bubbles

3(b)(iv)

- $4\text{OH}^- \rightarrow 2\text{H}_2\text{O} + \text{O}_2 + 4\text{e}^-$
- any negatively charged OH species losing electrons
- OR
- $\text{H}_2\text{O} + \text{O}_2$ as only products (other than electrons) on RHS
- any negatively charged OH species losing electrons

- AND
- $\text{H}_2\text{O} + \text{O}_2$ as only products (other than electrons) on RHS
- correct equation

3(c)(i)

- increases

3(c)(ii)

- no change

3(d)(i)

- bauxite

3(d)(ii)

- cryolite

3(d)(iii)

- anodes or carbon / graphite react with oxygen / O_2
- (form) carbon dioxide

Easy explanation / reminder

- *Electrolysis tip: cations go to the cathode and gain electrons. Anions go to the anode and lose electrons.*
- *Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.*

2025 Oct/Nov Paper 43 Question 3

Main topic: Topic 4: Electrochemistry | Marks: 17 | Answer source: 0620_w25_ms_43.pdf

Extra topic(s): Topic 9: Metals

Answer points

3(a)(i)

- aluminium oxide

3(a)(ii)

- cryolite

3(a)(iii)

- carbon

3(a)(iv)

- $\text{Al}^{3+} + 3\text{e}^- \rightarrow \text{Al}$
- only $\text{Al}^{3+} + 3\text{e}^-$ on the left
- equation completely correct

3(b)(i)

- unreactive coating of aluminium oxide

3(b)(ii)

- $2\text{Al} + 3\text{Zn}^{2+} \rightarrow 2\text{Al}^{3+} + 3\text{Zn}$

3(b)(iii)

- $\text{Zn}^{2+} = +2$
- $\text{Zn} = 0$ / zero

3(b)(iv)

- decrease in oxidation number

3(c)(i)

- enthalpy change (of reaction)

3(c)(ii)

- (forward) reaction is exothermic

3(d)

- oxides that react with acids AND with bases to produce a salt and water

3(e)

- $2\text{NaOH} + \text{Ga}_2\text{O}_3 \rightarrow 2\text{NaGaO}_2 + \text{H}_2\text{O}$
- NaGaO_2 on the right-hand side of an equation
- equation fully correct

3(f)

- GaBr_3
- $\text{Ga}_2(\text{SO}_4)_3$

Easy explanation / reminder

- *Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation before converting to the final answer.*
- *Electrolysis tip: cations go to the cathode and gain electrons. Anions go to the anode and lose electrons.*

